

LLM-Accelerated Augmentative and Alternative Communication for Urdu-Speaking Users with Severe Motor-Speech Impairment

Abstract

People with severe motor-speech impairment who cannot speak intelligibly rely on Augmentative and Alternative Communication (AAC), typically operated one slow physical selection at a time. Recent work shows that large language models can substantially cut this effort for English-speaking AAC users, but no published system extends this benefit to Urdu, or to any South Asian language, despite a large and severely underserved population. This proposal presents a tablet-based, LLM-accelerated AAC tile board for Urdu-speaking users, adapted specifically to Urdu’s morphology, right-to-left script, and routine code-mixing with English. The central research question is whether the motor-action-saving effect demonstrated for English AAC users transfers to a structurally different, low-resource language, and whether it holds under realistic code-mixed input. The proposal grounds this question in documented Pakistani clinical need, identifies concrete candidate recruitment sites, and lays out a within-subjects evaluation designed to produce a genuine, reportable empirical answer rather than simply a working prototype.

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Glossary

A short reference for clinical and disability-related terms used throughout this proposal, included since the intended readership includes people without a medical or speech-language-pathology background.

Amyotrophic lateral sclerosis (ALS)

A progressive neurological disease that gradually destroys the nerve cells controlling voluntary muscles, including those used for speech, while typically leaving cognition intact.

Anarthria A total or near-total loss of the physical ability to articulate speech. It is a motor problem, not a language or thinking problem; the person can still understand language and form thoughts normally.

Aphasia A language impairment, often caused by a stroke or brain injury, that affects the ability to produce or understand language. It can occur with otherwise normal cognitive function, and is distinct from a purely motor speech problem like dysarthria.

Augmentative and Alternative Communication (AAC)

Any tool or method, low-tech or digital, that helps a person communicate when they cannot rely on natural speech alone. Examples include picture boards, symbol grids, and on-screen keyboards.

Cerebral palsy

A group of permanent movement and posture disorders caused by a disturbance in the developing brain, usually before or around birth. It frequently affects the muscle control needed for clear speech.

Code-mixing The practice of using two languages together within a single sentence or conversation, common among bilingual speakers. In this proposal it refers specifically to Urdu and English used interchangeably in the same utterance.

Dysarthria A motor speech disorder in which the muscles used to produce speech are weakened, slowed, or hard to coordinate, making speech slurred or difficult to understand, even though the person's underlying language and cognitive ability may be unaffected.

Eye-gaze typing

A text-entry method that lets a user select letters or words on a screen by looking at them, tracked by a camera, used by people who cannot reliably use their hands for input.

MAKATON Pakistan

A Pakistani adaptation of Makaton, a sign- and symbol-supported communication system used alongside speech to support people with communication difficulties.

Motor-speech impairment

A general term for any condition that disrupts the physical production of speech, as opposed to language or cognition. Dysarthria and anarthria are both types of motor-speech impairment.

Picture Exchange Communication System (PECS)

A structured AAC method in which a user exchanges picture cards to request items or communicate needs, widely used in special education.

Speech-language pathologist

A clinician trained to assess and treat communication, language, and swallowing disorders, including AAC needs. Also referred to in this proposal as a speech-language professional.

TEACCH A structured-teaching approach originally developed for autistic individuals that uses visual schedules and organized environments to reduce cognitive and language demands during learning and communication.

1 Problem Statement

People with severe motor-speech impairment, including cerebral palsy with dysarthria or anarthria, amyotrophic lateral sclerosis, and post-stroke aphasia with preserved cognition, cannot speak intelligibly and rely instead on Augmentative and Alternative Communication (AAC): picture boards, symbol grids, or on-screen keyboards operated one slow physical selection at a time. This is a genuine bottleneck. Composing even a short sentence can take many selections, and the effort involved discourages spontaneous communication.

Recent work in English-speaking contexts shows that large language models can substantially reduce this burden by predicting what an AAC user is likely to want to say next. Cai et al. [1], in a study published in *Nature Communications*, report that an LLM-powered abbreviated-text-entry system reduced required motor actions by 57% in offline simulation, and produced text-entry rates 29 to 60% above baseline for two eye-gaze AAC users with amyotrophic lateral sclerosis in field testing. Valencia et al. [2], presenting at the 2023 CHI Conference on Human Factors in Computing Systems, studied 12 AAC users interacting with live LLM suggestions and found real gains in speed and expressiveness, alongside a documented risk: when suggestions are inserted without active review, users can feel that the resulting words are no longer their own.

This body of evidence is entirely English-language. No published system extends LLM-based phrase acceleration to Urdu, or, based on a literature scan conducted for this proposal, to any South Asian language. This is a consequential gap, not a minor one. Urdu is morphologically richer than English, is written right-to-left, and is routinely code-mixed with English within a single sentence by Pakistani speakers and their families. A predictive system built or evaluated only on English, or naively translated into formal monolingual Urdu, cannot be assumed to work, since none of the structural or sociolinguistic conditions that made the English result hold have been tested outside English. Whether the demonstrated motor-action-saving effect transfers to a language this different, and whether the documented “not my own voice” risk manifests differently in a different linguistic and caregiving context, are open empirical questions.

The scale of unmet need in the country this project focuses on is well documented and severe. A hospital-based survey of 225 children with cerebral palsy in Muzaffarabad, Azad Jammu and Kashmir, found speech impairment present in 70.2% of cases, and reported that 94.3% of the children with speech impairment had never consulted a speech therapist, with none having received any therapy sessions at all [4]. Existing local AAC practice, where it exists, is limited to low-tech physical communication boards and MAKATON Pakistan sign-supported communication, as documented in Pakistani rehabilitation case literature [3]. The population this project addresses is therefore not one already reasonably served by existing AAC infrastructure that a digital tool would incrementally improve. It is a population with almost no access to structured communication support at all, for whom the question of whether a modern predictive AAC interaction can be introduced, learned, and found useful is itself substantively new.

2 Related Work

2.1 LLM-Accelerated AAC (English-language)

Cai et al. [1] developed a system that expands short, low-effort abbreviations into full, contextually appropriate phrases using a large language model, evaluated through offline simulation and a field study with two eye-gaze-typing users with ALS. The core interaction pattern this proposal adapts is the general shape of that loop: a small number of low-effort user selections, expanded by a language model into a complete, context-appropriate sentence, which the user reviews before it is produced as output.

Valencia et al. [2] is the most directly relevant prior human-subjects study. It tested live LLM suggestions with 12 AAC users across three scenarios (extending short replies, answering biographical questions, requesting assistance) and is notable for reporting not only benefits but a specific, named risk: suggestions that are inserted or sent without deliberate user review can make communication feel less authentically the user’s own. This proposal treats that finding as a direct design constraint rather than background context: every suggestion in the proposed system must be actively selected and confirmed by the user

before it is spoken, and this design choice, along with the “own words” concern, is carried into the evaluation as a measured outcome, not just a stated intention.

Gaines and Vertanen [5], published in Findings of the Association for Computational Linguistics: EMNLP 2025, address a more technical but directly relevant problem: converting subword-level language model predictions into the character-level predictions many AAC systems require, and showing that adapting the model to a curated set of AAC-relevant conversational sentences, rather than using it as-is against generic web text, measurably improves prediction quality. This proposal adopts the same domain-adaptation principle for Urdu: prediction is grounded in a small, curated corpus of everyday Urdu AAC-relevant sentences, including Urdu-English code-mixed forms, rather than a generic Urdu corpus.

2.2 Interaction and Input Design Principles

Lohar et al. [6], published in Computers, demonstrate that a minimal hand-gesture vocabulary reduces cognitive load and response time for visually impaired users compared to a larger gesture set. The general principle, that input complexity should be kept as low as possible for users already managing significant physical or cognitive load, is applied here to the number and consistency of tile selections required to compose a message.

The Picture Exchange Communication System [7] and Structured TEACCHing [8] are long-established, well-evidenced special-education practices showing that many users with communication impairments engage more easily with structured picture, icon, or symbol systems than with open-ended text or speech, since visual supports lower working-memory and language demands. These form the basis for the icon-and-word-tile interaction layer proposed here, rather than a free-text or keyboard-first design.

2.3 Evaluation Stance for Disability-Focused HCI Work

Spiel and Gerling [9], in a critical review published in ACM Transactions on Computer-Human Interaction, argue that HCI research involving disabled and neurodivergent populations has too often framed the population as a deficit to be corrected, rather than centering the population’s own autonomy and self-directed use. This proposal adopts that stance explicitly: the system is designed and evaluated as a communication and expressiveness tool that participants may find useful, not as an intervention scored against a non-disabled speech baseline.

2.4 Local Clinical Context

A hospital-based survey in Muzaffarabad, Azad Jammu and Kashmir, of 225 children with cerebral palsy documents both the prevalence of speech impairment in this population (70.2%) and the near-total absence of speech-therapy access (94.3% never consulted, none receiving therapy) [4]. A separately published Pakistani clinical case report describes the use of a physical communication board and MAKATON Pakistan sign-supported communication with a child with cerebral-palsy-related dysarthria, confirming that structured AAC practice, in low-tech form, already exists within Pakistani clinical and special-education settings [3]. Together, these sources establish both the severity of unmet need and the existence of a real, existing clinical and educational context into which a digital tool would be introduced, rather than one requiring the population and practice to be constructed from nothing.

3 Research Gap and Contribution

No LLM-accelerated AAC system has been built or evaluated for Urdu, or, as far as this proposal’s literature review found, for any South Asian language. This gap is not merely geographic. Urdu differs from English in ways directly relevant to whether a phrase-prediction system will work at all: it is morphologically richer, uses a right-to-left script, and is routinely code-mixed with English within a single sentence in real Pakistani usage. A system and evaluation built to naively mirror the English approach, without testing against code-mixed input, would not actually answer whether the underlying effect generalizes.

This project’s central research question is therefore: does LLM-based phrase prediction reduce the physical selection effort required to compose a message for Urdu-speaking AAC users, in the way it has been shown to for English-speaking users, and does this hold when the input includes realistic Urdu-English code-mixing? A secondary question, following directly from Valencia et al. [2], is whether the “own words” authenticity concern documented for English AAC users manifests similarly, more, or less in this different linguistic and caregiving context.

These are testable, currently unanswered questions with a real possibility of a negative or mixed result, which would itself be a legitimate and reportable finding: for instance, if code-mixed prediction quality is substantially worse than monolingual prediction quality, or if the motor-action savings measured for English do not transfer at the same magnitude, both are informative outcomes for the literature on cross-linguistic generalization of AAC-LLM systems, not failures of the project. The contribution is not the existence of a working prototype by itself, but an empirical answer, grounded in real user data from a population with a documented and severe unmet need, to a question the existing literature has not yet addressed.

4 Proposed System

The proposed system is a tablet-based AAC communication board for Urdu-speaking users with severe motor-speech impairment. A user composes a message by selecting a small number of icon or word tiles representing common needs, people, feelings, and actions. After one to three tile selections, a language model, adapted against a curated Urdu AAC sentence corpus, suggests a small ranked set of complete, natural Urdu sentence candidates consistent with the selections made so far. The user reviews these candidates and taps to select one; no suggestion is spoken until it has been actively confirmed. The confirmed sentence is then spoken aloud using Urdu text-to-speech.

4.1 Core Components

- An icon-and-word-tile board covering common needs, people, feelings, and actions, each tile tied to an Urdu word or short phrase and touch-selectable.
- A phrase-prediction layer using an existing multilingual or Urdu-capable language model, adapted through prompting and light domain adaptation against a small, team-curated corpus of everyday Urdu AAC-relevant sentences, deliberately including Urdu-English code-mixed forms rather than only formal monolingual Urdu.
- A review-and-confirm interaction model in which every prediction is presented as a selectable option; the system never speaks or sends a suggestion without explicit user confirmation.
- Urdu text-to-speech output for the finished, confirmed message.
- A small per-user personalization layer allowing a small number of personally relevant names, foods, and phrases to be added to an individual’s board.

4.2 Scope Boundaries

The system does not attempt dysarthric or atypical speech recognition as an input method. Input is entirely tile and switch selection. This is a deliberate scoping decision: atypical Urdu speech has essentially no public dataset available, and attempting speech-based input would substitute a severe, likely infeasible data-collection problem for a well-scoped interaction and evaluation study. The system also does not attempt to train a new language model; it uses an existing model via prompting and lightweight domain adaptation, consistent with the approach demonstrated by Gaines and Vertanen [5].

5 Methodology

5.1 Participants and Recruitment

Given the prevalence and access data reported in Section 1, Pakistan has a real, identifiable population of children and adults with cerebral-palsy-related or similar motor-speech impairment. Within Pakistan, this project’s recruitment is scoped to Karachi, where several rehabilitation and special-education centers already serve this population. Recruitment for this project is planned through an existing rehabilitation or special-education center in Karachi rather than open public recruitment, which is both more realistic given the population’s needs and consistent with how AAC research is normally conducted through a clinical or educational intermediary. Candidate recruitment sites identified for this proposal, all based in Karachi, include Imran Rehabilitation Centre in Gulistan-e-Jauhar, which has focused on cerebral palsy rehabilitation for over sixteen years; Al Umeed Rehabilitation Association for Cerebral Palsy (AURA), also in Gulistan-e-Jauhar, which specializes specifically in cerebral palsy rehabilitation; and Al-Shifa Trust Special School and Centre for Rehabilitation in CAA Colony, which serves children with developmental disorders including cerebral palsy. A target sample of five to eight participants is planned, consistent with standard practice for within-subjects AAC usability studies, and is achievable by recruiting through a single center in Karachi.

Because the Muzaffarabad data [4] indicate that the large majority of children with speech impairment in Pakistan have never received speech therapy, most recruited participants are expected to be new to structured AAC rather than established users. The study design accounts for this directly rather than assuming AAC proficiency at intake.

Informed consent will be obtained from participants and, where applicable, from guardians or caregivers. A speech-language professional at the recruiting institution will be involved in task design and will supervise all sessions; no session will be conducted unsupervised.

5.2 Procedure

Each session begins with a short, supervised familiarization phase introducing the tile-board interaction, given that most participants will not already be experienced AAC users. Following familiarization, participants complete a set of tasks under two conditions in a within-subjects design: a baseline condition using a plain Urdu tile or word board with no language-model prediction, representing the digital equivalent of the low-tech boards currently in use, and the proposed condition using the language-model-assisted board.

Tasks include composing and speaking a greeting or need-based message from a small tile set, composing a message that requires Urdu-English code-mixed content, and composing a short reply to a simple question.

5.3 Measures

Quantitative measures are the number of selections required to complete each message, time to message completion, and task success rate. The selections-per-message measure is the direct Urdu-language analog of the “motor actions saved” metric reported by Cai et al. [1], allowing a direct, if informal, point of comparison to the English-language result.

Subjective measures are the System Usability Scale, the NASA Task Load Index, and a short item asking participants directly whether the message they composed felt like their own words, a direct empirical follow-up to the authenticity concern raised by Valencia et al. [2].

Qualitative data will be gathered through a short post-task interview or, where more appropriate given participant communication ability, a caregiver or therapist observation form, thematically coded for themes including handling of code-mixed input, trust in the system’s suggestions, and whether the available tile vocabulary matched participants’ real communication needs.

6 Risks and Limitations

This study involves a vulnerable population whose communication impairment can itself complicate the process of obtaining direct informed consent. This is addressed through guardian consent where applicable, involvement of a speech-language professional at the recruiting institution throughout task design and session supervision, and a firm rule against any unsupervised session.

General-purpose Urdu language model quality, particularly on code-mixed and colloquial text, is less mature than for English. This is treated in this proposal as a testable risk rather than an assumption: the evaluation deliberately includes code-mixed tasks specifically to surface this limitation empirically rather than avoid it.

A sample of five to eight participants will produce findings that are indicative for the specific population and setting studied, not generalizable to all Urdu-speaking AAC users, and this limitation will be stated explicitly in any resulting report.

Because most participants are expected to be new to structured AAC rather than established users, the comparison this study can support is specifically whether language-model-assisted tile selection is easier to learn and use than a static board for a first-time or early-stage AAC user. It cannot, on its own, establish how the system compares to years of established AAC practice, and this scope will be stated explicitly.

AAC message content can include personal or sensitive material. Any use of a cloud-hosted language model will be disclosed plainly to participants and guardians as part of consent, and an on-device or locally hosted model will be used in preference to a cloud API where feasible.

Finally, this project is evaluated according to the stance argued by Spiel and Gerling [9]: as a communication and autonomy tool that participants may find useful and expressive, not as a corrective intervention measured against non-disabled speech, and reporting will avoid deficit framing throughout.

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